AGRI ORACLE

AI Powered Solution for Crop Disease Detection and Crop Rotation

Paramjeet Singh
Department of Computer Science Engineering
VIPS-TC
New Delhi, Delhi
paramjeetsingh.21022004@gmail.com

Rachit Garg
Department of Computer Science Engineering
VIPS-TC
New Delhi, Delhi
rachitgarg1208@gmail.com

Rishab Sharma
Department of Computer Science Engineering
VIPS-TC
New Delhi, Delhi
rishabsha05@gmail.com

Shivanka
Department of Computer Science Engineering
VIPS-TC
New Delhi, Delhi

Abstract

Crop health monitoring the process of assessing crops for diseases and pests that affect the crop yield and quality. Traditionally, farmers have relied on their knowledge and experience to tackle any obstructions in the farming process, but this method is highly inefficient and limited to the knowledge of the farmer. With changing times and evolving machine learning algorithms, we wish to do away with the traditional way of detecting crop diseases and shift towards a smart and comprehensive method. Our methodology for disease detection includes utilizing MobileNetV2 for deep learning classification. Our dataset consists of images of crops such as Cotton, Potato, Rice, Tomato and Wheat. By dividing our dataset into healthy and diseased crops, we train the model to pick up on visual cues, indicating disease. For crop rotation model, Multi Class Classification was leveraged to suggest a crop after taking inputs about previous crop, soil type and sowing season. These inputs help us recommend the farmer a certain crop which will ensure that the soil isn't depleted of its nutrients while also allowing the farmer to continue cultivation.

Keywords: Agriculture, MobileNetV2, Machine Learning, Transfer Learning, Multi-Class Classification

1. Introduction

India is a major agricultural producer which employs about 45% of the population. According to estimates, 18% of country's GDP is generated from the agricultural sector and we are the largest producer of pulses and second largest producers of rice and wheat. Green revolution of mid 1960s lead to major increases in the crop yield which also helped to alleviate the challenges of starvation and poverty in the country [1]. This change was brought on by adopting to use GMO seeds, fertilizers and pesticides to increase crop production and minimize losses due to due to pest infestations and soil depletion. But due to heavy usage of pesticides and artificial fertilizers have led to drastic decrease in soil quality thus leading to the same problem it once solved. We also encounter profound environmental challenges due to conventional agriculture methodologies. A technological disruption can greatly benefit the country's agricultural output while also mitigating a lot of challenges faced by the farmers. In India pests and diseases lead to an annual loss of INR 290 billion. In 2012 alone, fungal diseases ruined around125 million tons of essential crops like wheat, rice, soybeans, maize, and potatoes [2]. With timely detection and appropriate treatment these losses could have been easily averted.

In this age of evolving technology, we wish to revolutionize the way farmers seek remedies for their crop related problems by making the process more efficient and technology-driven. By simply snapping a photo of their crop and uploading it to the website, the ML model will scan through its dataset for various fungal or bacterial diseases and show the appropriate output. In addition to crop disease detection, our platform has integrated a crop rotation model that suggests optimal choice of crop to be planted based on the previous cultivation history of the farmer. Crop rotation with diversification of species is a suitable management of pests, diseases and weeds and improves soil fertility, which can lead to profitability enhancement in grain production system [3]. Our project aims to design a methodology for image processing for a diverse range of crops and their related diseases with better metrics by applying the lightweight deep learning architecture of MobileNetV2. By building a model with good precision and accuracy along with a variety of dataset, we contribute towards the advancement of technology in the agricultural sector.

2. Literature review

With recent technological advancements, particularly in the field of computer visionmany researchers have utilized CNN for disease detection in crops due its ability to pick up on visual cues and correctly predict with a high accuracy. This section aims to provide an insight into existing metholdogies and approaches used by researchers who leveraged CNN in crop disease detection.

The approach taken by Rani et al. [4] was to use transfer learning on the pre-trained deep learning model of VGG16 or ResNet. Data augmentation was utilized for a model that could generalise well in unfavourable image conditions such as low lighting, flipped images, skewed photos etc. CNN model to detect diseases in Bell Paper, Tomato and Potato crops was proposed by Garima Shreshta et al. [5]. The model proposed achieved a training accuracy of 97.42% while test accuracy was 88.80%. However, the high accuracy number doesn't depict the working of the model, for instance, classes 0, 1, and 2 show a relatively higher precision and recall, but others, such as class 3, 5, and 6, have precision and recall scores of 0.00, indicating that the model fails to identify those diseases in most cases. Low macro-average and weighted-average scores also shows model's inability to recognize certain classes. Relying solely on accuracy can be misleading, as it does not capture class-wise performance or the model's ability to generalize. This highlight the importance of reviewing important performance metrics to determine model's suitability for practical use. S. A. Upadhye, M. R. Dhanvijay, and S. M. Patil [6] utilized Convolutional Neural Networks (CNNs) to identify various diseases in sugarcane and their model achieved a classification accuracy of 98.69% across four categories. They also provide remedies for the disease which the model detects. Kunduracioğlu, Ismail & Pacal, Ishak [7] tested out 13 different models for sugarcane dataset which contained 11 classes. Their findings lead to the conclusion that EfficientNet b6 and Inception V4 were able to generate the highest accuracy scores of 93.39% and 93.10% respectively. The dataset consisted of 6748 images. Emma Harte [8] deployed the CNN model in a web application format which was designed to classify an image into one of the seven possible classes. The dataset used contained of 8685 images of potato, tomato and rice and their related diseases. The ResNet34 model used achieved an accuracy of 97.2%. These accuracy figures demonstrate the model's ability to generalise and emphasized the practical ability. The leaf spot disease on sugar beet plant were discussed by Ozguven et al. [9] and their methodology was to utilize and Updated Faster R-CNN. The dataset for the crops includes 155 images for training and testing. Their proposed model achieved an accuracy of 95.48%. Militante et al. [10] created a CNN model with 96.5% accuracy. They identified and detected 32 plant diseases and species with 96.5% accuracy. Real-time images were processed for plant disease detection and identification using the trained model. Their proposed research has vast potential in assisting farmers to identify plan health problems with accuracy. Punam Bedi and Pushkar Gole [11] A hybrid model of CAE-CNN was used in the detection of Bacterial Spot disease in peach plants in this research. The model was 99.35% accurate when trained and 98.38% accurate when tested. An open dataset in PlantVillage was employed. R. Akter and M. I. Hosen [12] analysed 34123 images belonging to ten different Bangladeshi medicinal plants for leaf recognition. A CNN model was used and the results based on both training and testing set came out to be 71.3 %. Vaibhav Tiwari et al. [13] propose a deep learning-based system using dense CNN architectures to detect and classify plant diseases across six crops in 27 categories. The model achieved 99.58% accuracy and 98.199% on unseen test images. With 0.016 second processing time for each image, the model performs well in real-time.

3. Methodology

3.1 Crop Disease Detection

The proposed machine learning model asks for an input from the end user (farmer) in the form of an image. Then the model recognises and picks up on certain visual cues which helps it classify the image into one of the seventeen possible classes. The flow of the ML model is represenetd by the Fig.1. Firstly the image dataset is loaded using OpenCV before converting them into NumPy arrays using ImageDataGenerator. Next, preprocessing is applied which involves resizing the images and applying data augmention such as rotational range, brightness range, horizontal flip and shear range. The labels of the images are then extracted from the subfolders before splitting the data into training and testing set. For this iteration of our model, we decided to use 25% of the dataset for testing. After importing the pre-trained model of MobileNetV2, the last 25 layers were kept for fine-tuning the pre-built architecture. This approach of transfer learning allows for efficient learning and faster convergance.

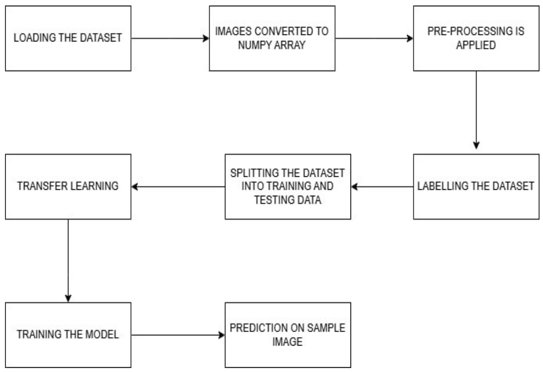

Fig. 1. Block diagram depicting flow of ML model

For a given image, there are 20 distinctive probaility values for the 20 classes respectively. The label with the highest probaility score will be identfified as the predicted disease outcome for the given image.

3.2 Crop rotation

The dataset was created with information on previous crops, soil type and the sowing season. These categorical variables are converted to numerical representation using One-HotEncoder. Considering all these inputs, the model will recommend a next crop with a reasoning for the same. A Sequential Deep Learning Model was implemented using TensorFlow/Keras. The architecure includes two dense layers of 256 neurons. The model suggest the next crop bases on these factors:

- Previous crop- currently the model predicts for only Rice, Wheat, Potato, Tomato, Corn, Barley, Peas, Soybean, Sugarcane, Cotton, Groundnut, Millet, and Maize
- Soil type- clay, loamy or sandy
- Growing season- Kharif or Rabi

4. MOBILENETV2

MobileNetV2 is a lightweight Convoultion Neural Network archirture that was developed by Google. It was an improvement over its predeccesor MobileNetV1 and foucses on reduing computional costs while manintaing a high accuracy. This was achieved by utilizing residuals and lineaer bottlenecks.

The MobileNetV2 architecture is built upon depthwise separable convolutions (like MobileNetV1) and it is described as below:Initial Convolutional Layer

The network starts with a 3×3 standard convolutional layer containing 32 filters and a stride of 1. This layer extracts the basic features such as edges and textures from the input image. Batch normalization is implemented to stabilize training .

4.1 Inverted Residuals

MobileNetV2 uses inverted residuals unlike traditional residual connections. In inverted residuals expansion layers first increase the number of channels before applying depthwise convolutions.

This method allows for efficient feature extraction while reducing the number of parameters

4.2 Linear Bottlenecks

Instead of utilizing ReLu in the final bottleneck layer, MobileNetV2 instead uses a linear activation function to retain important low dimensional feature and prevent information loss.

4.3 Final Convolutional Layer

A 1×1 convolution with 1280 filters is applied towards the end of the network, which further refines the extracted features. To compress every feature map to a single numerical value by averaging all its spatial elements, the global average pooling layer is implemented. This also guarantees that the final representation would be compact yet informative for the purpose of classification.

4.4 Fully Connected Layer and Output

The last layer in MobileNetV2 is a fully connected layer that projects the processed features to the target number of output classes. A softmax activation function is then used to generate probability scores for every class, allowing the network to make its final classification choice.

4.5 Transfer Learning

It refers to the techqniue of leveraging existing models rather than buidling a model from scratch. This pre-trained model is time efficient and reduces the complexity of the model leading to better generalisation and hence better results. We have utilized MobileNetV2 for our model and we changed the last 25 layers of the pre-built model. By adding L2 regularisation which helps in preventing overfitting and adjusting the learning rate, we achieved a high training accuracy to high validation accuracy score.

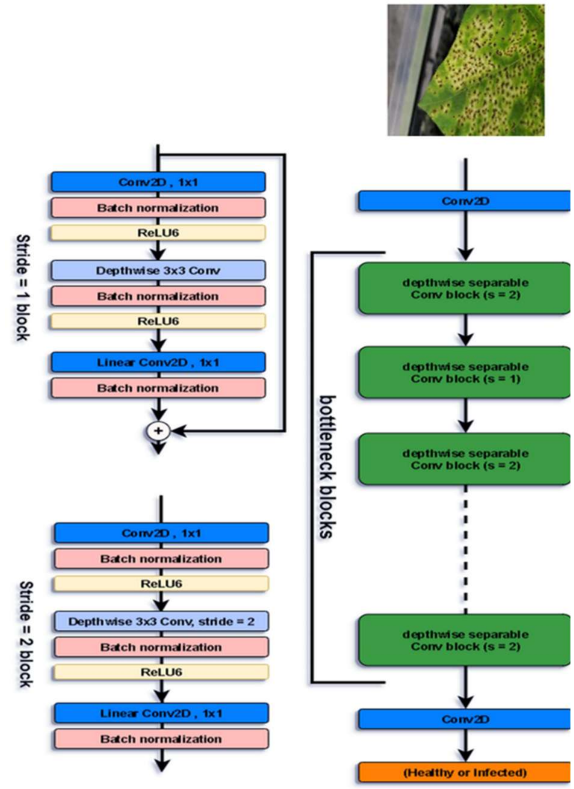

Fig. 2. Architecture of MobileNetV2 with a sample image [14]

Fig. 3. Sample images from the database (from top left to top right) Wheat Smut, Cotton Leaf Mildew

(from bottom left to bottom right) Rice Scald, Cotton Bacterial Blight

4.6 Global Average Pooling

Global Average Pooling (GAP) is a commonly used down sampling technique, used typically in Convolutional Neural Networks (CNN) prior to the connected (FC) layer. GAP downscales the spatial size of a feature map to a value by computing the mean of a feature map. Fig. 4 demonstrates the working of GAP with an example.

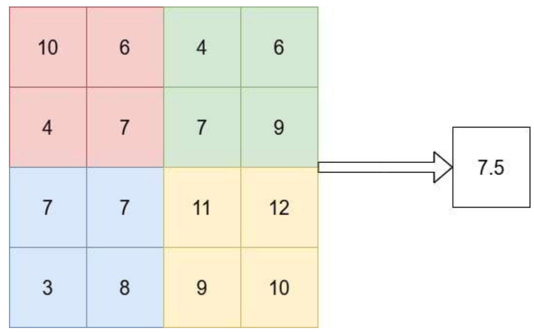

Fig. 4. Global Average Pooling

5. Multi-class Classification

Multi-class classification is a form of machine learning where the aim is to classify into more than two different changes compared to binary classification that only has two classes: yes or no. In the case of agricultural systems in particular, crop rotation modelling multi-class classification has a very important role to play in recommending or predicting the most suitable crop to cultivate in the next season, based on inputs entered by the farmer (previous crop, soil type and next season).

The multi-class classification model treats the crop recommendation task as a mapping function f(X)=Y

where:

- X refer to a set of input features (e.g., soil type, temperature, previous crops, rainfall)
- Y is one of the multiple crop categories (e.g., wheat, maize, rice, legumes)

6. Web Interface

React is a JavaScript library used to build user interfaces, especially for websites and web apps that change often, like social media platforms, dashboards, or online stores. It was created by Facebook and is now used widely by developers. React has the main feature called components which are like building blocks of a webpage. For example, a button, a menu, or a search bar can all be components and each component has its own design and behavior. React uses JSX, which is HTML mixed with JavaScript. This makes it easier to write the layout of website by writing HTML and JavaScript in a single file. React is based on virtual DOM. The DOM (Document Object Model) is what browsers use to show content on the screen. React uses a virtual DOM, which is like a copy. When something changes, React compares the new virtual DOM to the old one and only changes and updates the parts that are changed.

Flask is a framework made with help of Python that helps build websites and web apps. It's called a web framework, and it gives the basic parts needed to make an app work on the internet. Flask is also known as a micro framework. It means Flask is small and simple. It doesn't come with too many built-in features. One of the most important things Flask does is handle routes which is a web address that shows something when people visit it.

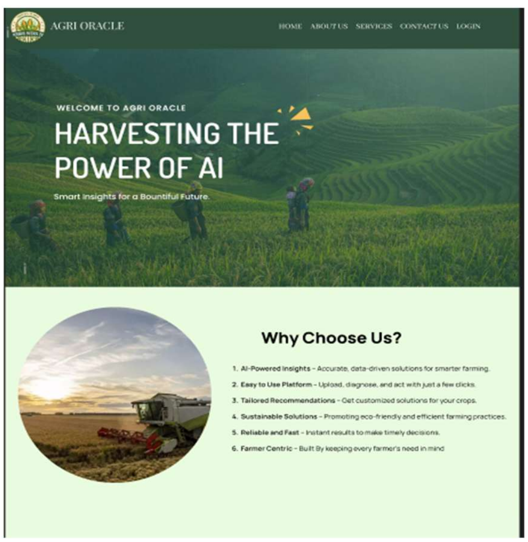

Fig. 5. Website Landing page

Fig. 6. Webpage highlighting our services

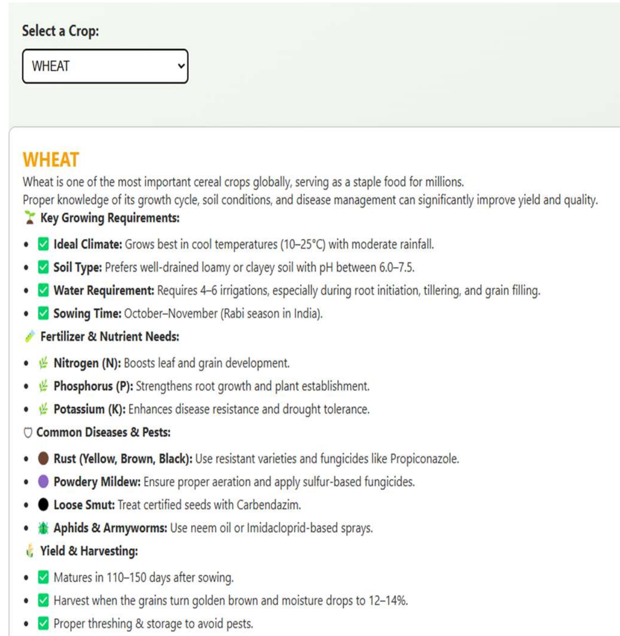

Fig. 7. General information webpage

7. Result

The collection of images in excess of 3000 is split by a ratio of 75:25 for test set and training set respectively. Overall training duration of the model was around 30 minutes once after which metrics of performance were observed. A training accuracy of 94.7% and validation accuracy of 86.82% was achieved. The slim difference in the two accuracy figures proves that the model doesn't suffer from overfitting. Fig. 8 depicts the Accuracy vs Epoch graph; Fig. 9 depicts the confusion matrix and Fig. 10 is the prediction made by the model on sample image fed to it. Fig. 11 is the classification report generated after the training. It includes metrics such as precision, recall and F1 scores. These metrics are crucial to understand the inner workings of the model despite the accuracy figures. To compare the performance of the transfer learning model, we designed a traditional CNN based model and analyzed its training time, training accuracy and validation accuracy. The detailed comparison is depicted in Fig. 12. The working of crop rotation model in the web interface is shown in Fig. 13.

The system on which the model was trained features 24 GB RAM, an AMD Ryzen 7 7435HS processor (3.10 GHz), and an NVIDIA GeForce RTX 4050 GPU with 6GB GDDR6.

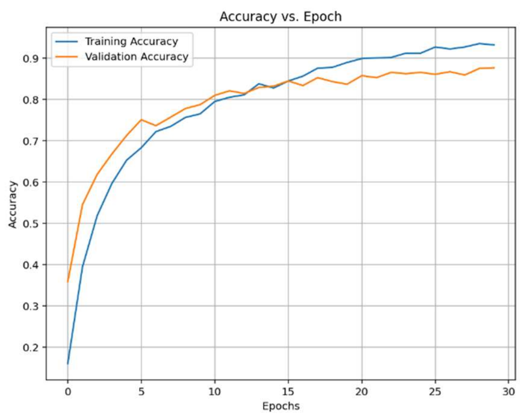

Fig. 8. Accuracy vs Epoch graph (Crop disease detection model)

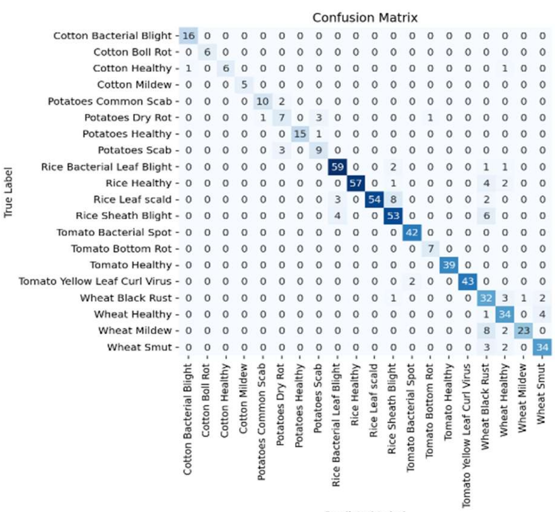

Fig. 9. Confusion Matrix (Crop disease detection model)

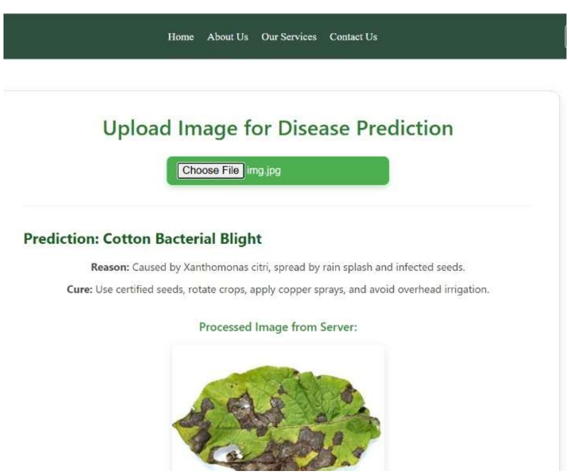

Fig. 10. Predictions made by the model

Class Name	Recall	Precision	F1 Score	Support
Cotton Bacterial Blight	0.94	1.00	0.97	16
Cotton Boll Rot	1.00	1.00	1.00	6
Cotton Healthy	1.00	0.75	0.86	8
Cotton Mildew	1.00	1.00	1.00	5
Potato Common Scab	0.91	0.83	0.87	12
Potato Dry Rot	0.58	0.58	0.58	12
Potato Healthy	1.00	0.94	0.97	16
Potato Scab	0.69	0.75	0.72	12
Rice Leaf Blight	0.89	0.94	0.91	63
Rice Healthy	1.00	0.89	0.94	64
Rice Leaf Scald	1.00	0.81	0.89	67
Rice Sheath Blight	0.82	0.84	0.83	63
Tomato Bacterial Spot	0.95	1.00	0.98	42
Tomato Bottom Rot	0.88	1.00	0.93	7
Tomato Healthy	1.00	1.00	1.00	39
Tomato Yellow Leaf Curl	1.00	0.96	0.98	45
Wheat Black Rust	0.56	0.82	0.67	39
Wheat Healthy	0.76	0.87	0.81	39
Wheat Mildew	0.96	0.70	0.81	33
Wheat Smut	0.85	0.87	0.86	39
Macro avg.	0.89	0.88	0.88	627
Weighted avg.	0.90	0.88	0.88	627

Fig. 11. Classification Report

Model type	Training time	Training Accuracy	Validation Accuracy
Traditional CNN	Approx. 40 minutes	88.1%	79.92%
Transfer Learning	Approx. 30 minutes	94.7%	86.82%

Fig. 12. Comparison between two models

From the figure 12, we can see a signifcant difference in all aspects of the model. A shorter training time along with higher accuracy was the outcome desired by leveraging Pre-trained MobileNetV2 model. This comparison proves the hypothesis right.

Fig. 13. Crop Rotation model suggestion

8. Conclusion and Future Scope

We achieved a high training accuracy 94.7% and validation accuracy 86.82%, which is on par of many similar ML models with the added benefit of having a vaster crop and their related disease dataset. The classification report suggests that the model generalizes well for almost all of the classes. The only classes that are underperforming are Potato Dry Rot and Wheat Black Rust while there is room for improvement in Potato Scab too. The crop rotation model is suggesting the right crops and giving the correct reasoning behind the suggestion. The website is integrated with the model perfectly using Flask framework of Python. All the webpages are cross-integrated perfectly making each button click responsive. AGRI ORACLE is fully developed and ready for practical, real-world use by the farmers.

While this iteration of our proposed method shows a strong performance, there are a lot of areas that can benefit from improvements and new ideas that can be integrated to make AGRI ORACLE truly a one stop solution that farmers can safely rely upon. With technology evolving at a rapid pace, it is crucial to be updated with the recent technological trends that will aid a new and improved iteration of AGRI ORACLE. Some of these improvements that can be applied in the near future are:

- Expand Disease Dataset

Incorporate a broader dataset that includes more crop types and a wider variety of disease types. This will help the model generalize better across diverse farming scenarios.

- Explore Alternative Pre-Trained Models

To possibly improve performance and attain better generalization with higher validation accuracy, try out different effective pre-trained models like EfficientNet, DenseNet, or InceptionV3.

- Integrate Weather-Based Farming Guidelines

Develop and integrate a weather advisory system that provides real-time, location-specific farming guidance. This feature will help farmers adapt to unpredictable weather conditions and make informed decisions.

- Enrich Crop Rotation Dataset

Augment the dataset used for the crop rotation model to include a wider range of crops, soil types, and account for regional farming practices. This will improve the accuracy and relevance of rotation recommendations.

References

[1] A. R. Eliazer Nelson, K. Ravichandran, and U. Antony, “The impact of the Green Revolution on indigenous crops of India,” Journal of Ethnic Foods, vol. 6, no. 1, Oct. 2019. doi:10.1186/s42779-019-0011-9

[2] S. Shukla, D. Upadhyay, A. Mishra, T. Jindal, and K. Shukla, “Challenges faced by farmers in crops production due to fungal pathogens and their effect on Indian economy,” Fungal Biology, pp. 495–505, 2022. doi:10.1007/978-981-16-8877-5_24

[3] B. Volsi, G. E. Higashi, I. Bordin, and T. S. Telles, “The diversification of species in crop rotation increases the profitability of grain

production systems," Scientific Reports, vol. 12, no. 1, Nov. 2022. doi:10.1038/s41598-022-23718-4

[4] Rani, S. S., Kumar, C. M. S., Felicita, S. A., Ganesh, S. S., Choubey, A., & Anitha, R. (2024). Development and Evaluation of a Distinctive Cloud-Based Artificial Intelligence System using Deep Learning Techniques (AISDLT) for Accurate Detection of Tomato Plant Leaf Diseases. IJISAE, 12(12s), 538–552.

[5] G. Shrestha, Deepsikha, M. Das, and N. Dey, "Plant disease detection using CNN," 2020 IEEE Applied Signal Processing Conference (ASPCON), pp.109–113,Oct.2020. doi:10.1109/aspcon49795.2020.9276722

[6] S. A. Upadhye, M. R. Dhanvijay and S. M. Patil, "Sugarcane Disease Detection Using CNN-Deep Learning Method," 2022 Fourth International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT), Mandya, India, 2022, pp. 1-8

[7] Kunduracioğlu, Ismail & Pacal, Ishak. (2024). Deep Learning-Based Disease Detection in Sugarcane Leaves: Evaluating EfficientNet Models. Journal of Operations Intelligence. 2. 321-235. 10.31181/jopi21202423.

[8] Harte, Emma. (2020). Plant Disease Detection using CNN. 10.13140/RG.2.2.36485.99048.

[9] Mehmet Metin Ozguvena, and Kemal Adem, "Automatic detection and classification of leaf spot disease in sugar beet using deep learning algorithms". Physica A: Statistical Mechanics and its Applications, Volume 535, 122537, 1 December 2019.https://doi.org/10.1016/j.physa.2019.1225 37.

[10] Sammy V. Militante, Bobby D. Gerardo, and Nanette V., "Plant Leaf Detection and Disease Recognition using Deep Learning," 2019 Eurasia Conference on IOT, Communication and Engineering (ECICE), Yunlin, Taiwan, pp. 579–582, Oct.2019 doi: 10.1109/ECICE47484.2019.8942686.

[11] P. Bedi and P. Gole, "Plant disease detection using hybrid model based on convolutional autoencoder and Convolutional Neural Network," *Artificial Intelligence in Agriculture*, vol. 5, pp. 90–101, 2021. doi:10.1016/j.aiia.2021.05.002.

[12] R. Akter and M. I. Hosen, "CNN-based leaf image classification for Bangladeshi medicinal plant recognition," *2020 Emerging Technology in Computing, Communication and Electronics (ETCCE)*, pp.1–6, Dec.2020. doi:10.1109/etcce51779.2020.9350900

[13] V. Tiwari, R. C. Joshi, and M. K. Dutta, "Dense convolutional neural networks based multiclass plant disease detection and classification using leaf images," *Ecological Informatics*, vol. 63, p. 101289, Jul. 2021. doi:10.1016/j.ecoinf.2021.101289

[14] MobileNetV2: Inverted residuals and linear bottlenecks|https://www.researchgate.net/publication/329740282_MobileNetV2_Inverted_Residuals_and_Linear_Bottlenecks (accessed Mar. 2, 2025).